UNIT-3

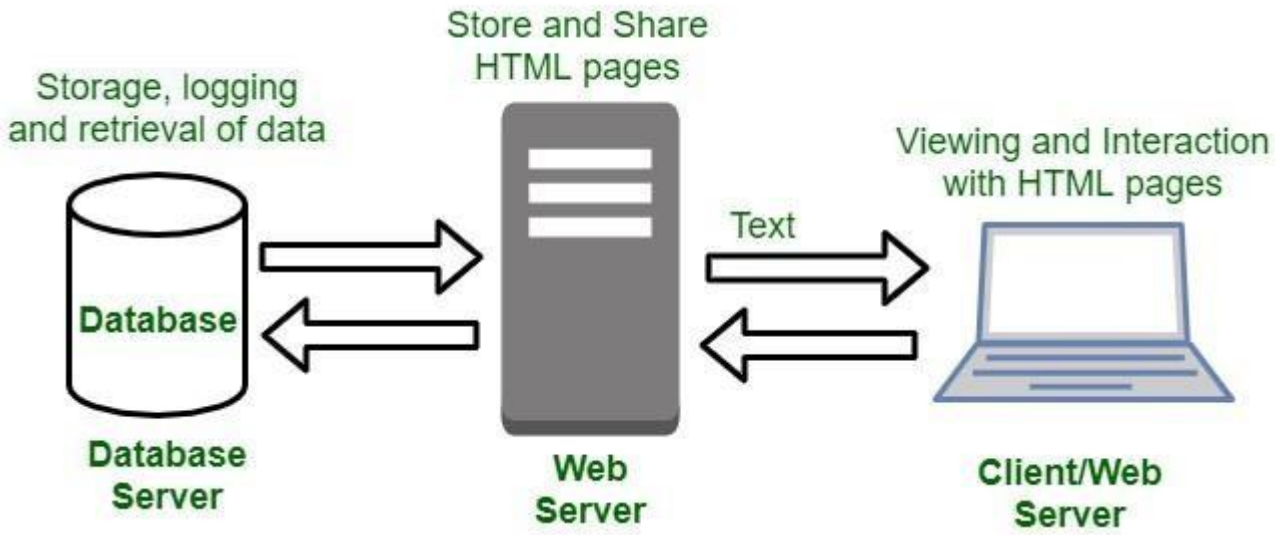
Planning and developing dynamic web content and Blog sites

Dynamic web pages

Dynamic Website is a website containing data that can be mutable or changeable. It uses client-side or server scripting to generate mutable content. Like a static website, it also contains HTML data.
Dynamic websites are those websites that changes the content or layout with every request to the webserver. These websites have the capability of producing different content for different visitors from the same source code file. There are two kinds of dynamic web pages i.e. client-side scripting and server-side scripting. The client-side web pages changes according to your activity on the web page. On the server-side, web pages are changed whenever a web page is loaded.

Example: login & signup pages, application & submission forms, inquiry and shopping cart pages.

A Typical Architecture of dynamic website



There are different languages used to create dynamic web pages like PHP, ASP, .NET and JSP. Whenever a dynamic page loads in browser, it requests the database to give information depending upon user's input. On receiving information from the database, the resulting web page is applied to the user after applying the styling codes.

Features of dynamic webpage:

- These websites are very flexible.
- In these websites the content can be quickly changed on the user's computer without new page request to the web browser.

- In these websites the owner have the ability to simply update and add new content to the site.
- These websites are featured with content management system, e-commerce system and intranet or extranet facilities.
- Most of the dynamic web content, is assembled on the web using server-scripting languages.

Advantages of dynamic webpage:

- It provides more functional websites.
- It is very easy to update.
- It helps in the search engines because new content brings people back to the site.
- These are interactive websites because these can be customized.
- These websites can work as a system to allow staff or users to collaborate. It's easy to modify or update data.
- It provides a user-friendly interactive interface for users.
- proves smooth navigation.

Static web pages

Definition: Static web pages are fixed and display the same content to every user. They are written in HTML and CSS and don't involve server-side processing.

Characteristics of static web pages:

Content remains constant unless manually updated by a web developer.
Typically faster to load because the server simply serves the pre-existing HTML files without any processing.

Suitable for websites with relatively stable content, such as informational sites or portfolios. Examples include basic personal websites, company brochures, or simple blogs.

Key Differences between static and dynamic web pages:

Content: Static pages display fixed content, while dynamic pages can generate content onthe-fly based on various factors.

Interactivity: Dynamic pages can offer interactive features, such as user authentication, form submissions, or personalized content, which static pages typically lack.

Development Complexity: Dynamic pages usually require more complex development with server-side scripting and database integration compared to static pages.

Performance: Static pages tend to load faster since they are served directly by the server without any processing, whereas dynamic pages may have additional processing overhead, depending on the complexity of the server-side logic.

Maintenance: Static pages are generally easier to maintain since their content doesn't change frequently, while dynamic pages may require more frequent updates and monitoring due to their interactive nature.

Dynamic Website Types

Depending on the scripting language, there are two types of dynamic websites.

Client-Side Scripting
A client-side scripting dynamic website changes its content in response to user actions on the page, such as clicking a button. Generally, client-side scripted website content is loaded in the visitor's web browser instead of the server. This script determines what user interactions trigger the event and which page content changes. When building this type of dynamic website, commonly-used client-side scripting languages include JavaScript and VBScript. As it doesn't use any server-side scripting languages, a dynamic site can be hosted statically.

Server-Side Scripting
Server-side scripted web pages' content is rendered on the host and changes before being displayed. This script instructs the application server on how to assemble the requested page.
Users' requests and inputs on the website can trigger the script. For instance, when users log in to their ecommerce accounts, the script will instruct the server to show their cart items.
Server-side scripts are used to build an interactive website. They let users enter data into forms or upload images, which will be stored in a database.

Planning and developing dynamic web content sites involves following steps:

1. Define Objectives: Clearly outline the purpose and goals of your website. Identify the target audience and their needs. Conduct market research to understand the audience's needs, competitors, and industry trends.

2. Content Strategy: Plan the types of content to be included, such as images, text, videos, or interactive elements, and how they will be organized and presented.

3. User Experience (UX) Design: Create wireframes and prototypes to visualize the layout, navigation, and user interface of the website. Consider user flows and interactions for a seamless experience.

4. Choose a Technology stack: Select the appropriate technology stack based on your requirements and expertise. Build the website using HTML, CSS, and JavaScript for front-end development, and PHP, Python, or Ruby for back-end development to handle dynamic content.

- **Frontend Development:** Build the frontend of the website using HTML, CSS, and JavaScript frameworks/libraries (e.g., Bootstrap, React, Vue.js).
- **Backend Development:** Develop the backend functionality of the website to handle dynamic content generation, user authentication, database

5. Dynamic Content Types: Incorporate dynamic features such as user-generated content, personalized recommendations, social media integration, and real-time updates to enhance user engagement and interaction.

6. Data Architecture: Plan the structure of your data to support dynamic content. Consider databases, data models, and APIs for data retrieval. Design and implement the database schema to store and manage dynamic content, user data, and other relevant information.

7. **Content Management System (CMS):** Choose a CMS that supports dynamic content creation and management. Ensure it allows for easy updates and modifications.

8. Testing: Conduct thorough testing of the website to identify and fix any bugs, usability issues, or performance bottlenecks. Test across different browsers, devices, and network conditions to ensure compatibility and responsiveness.

9. Deployment: Deploy the website to a production server or hosting platform. Configure server settings, domain name, SSL certificate, and security measures to ensure the website is accessible and secure.

10.Content Updates and Maintenance: Continuously update and maintain the website's content to keep it fresh, relevant, and engaging. Monitor user feedback, analytics data, and industry trends to identify opportunities for improvement and optimization.

11.Responsive Design: Implement a responsive design to ensure your dynamic content adapts to different devices and screen sizes.

12.Performance Optimization: Optimize code, images, and other assets to ensure fast loading times. Consider content delivery networks (CDNs) for improved performance.

13.Security Updates: Regularly update the website's software, plugins, and dependencies to patch security vulnerabilities and protect against potential threats.

14.Scalability: Design the architecture with scalability in mind to handle potential increases in traffic and data. Consider cloud-based solutions for scalability and flexibility.

Website Design using CSS

Definition: CSS is the acronym of “Cascading Style Sheets”. CSS is a language for specifying how documents are presented to users — how they are styled, laid out, etc. It provides an additional feature to HTML. It is generally used with HTML to change the style of web pages and user interfaces

Understand selectors, properties, and values:

Selector

A selector is used to target the HTML element(s) you want to style. There are various types of selectors in CSS, including:

Type selectors (also known as tag or element selectors): Target elements by their tag name. For example, p selects all <p> (paragraph) elements.

Class selectors: Target elements by their class attribute. The class selector is prefixed with a period ., for example, .className targets all elements with class="className".

ID selectors: Target an element by its id attribute. The ID selector is prefixed with a hash #, for example, #header targets the element with id="header". IDs should be unique within a page.

Attribute selectors: Target elements based on the presence or value of their attributes, for example, input[type="text"] targets all <input> elements with a type attribute value of "text".

Pseudo-classes and pseudo-elements: Target elements in a specific state or target specific parts of an element. For example, :hover applies styles when the user hovers over an element, and ::before is used to insert content before the content of an element.

Combinatory and multiple selectors: You can combine multiple selectors or use combinators (like , >, +, ~) to target elements based on their relationships or to apply a single set of styles to multiple selectors by separating each selector with a comma.

Property

A property is a type of attribute you want to style in an element, such as color, width, height, margin, background, etc. CSS properties are the qualities of elements you can manipulate.

Value

Values are assigned to properties to specify how you want to style the property. For example, if you're setting the color property, the value could be a color name (red), a hexadecimal value (#ff0000), an RGB value (rgb(255, 0, 0)), or even an HSL value (hsl(0, 100%, 50%)).

Syntax of CSS

The general syntax of a CSS rule consists of a selector followed by a declaration block. The declaration block contains one or more declarations separated by semicolons. Each declaration includes a CSS property and a value, separated by a colon:

```
selector
{
property: value;
}
```

Example

```
 p
{
 color: blue;
font-size: 16px;
}
```

In this example, the selector is p, targeting all <p> elements. There are two declarations within the declaration block. The first declaration sets the color property to blue, and the second sets the font-size property to 16px.

By mastering the use of selectors, properties, and values in CSS, you can control the presentation of web pages with precision and creativity.

```
<!DOCTYPE html>
<html>
<head>
   <title>Understanding CSS</title>
   <style>
      /* CSS code */
p {
color: blue;
font-size: 16px;
      }
   </style>
</head>
<body>
   <p>This is a paragraph styled using CSS.</p>
</body>
</html>
```

Responsive Design:

Implement responsive design techniques: Use media queries to apply different styles based on the characteristics of the device, such as screen size, resolution, and orientation**.**

Ensure your website adapts: Design your website to be fluid and adaptable, ensuring it looks and functions well on various devices, including desktops, laptops, tablets, and smartphones.

```
@media only screen and (max-width: 600px) {
p {
      font-size: 14px;
   }
}
```

Layout Design:

Create flexible layouts: Utilize CSS grid or flexbox to create responsive and visually appealing layouts that adjust based on screen size and content.

Consider frameworks: Frameworks like Bootstrap provide pre-designed layout components and grid systems that can expedite the layout design process.

```
.container {     display: grid;

grid-template-columns: 1fr 1fr;

grid-gap: 20px;

}
```

Typography:

Choose appropriate fonts: Select fonts that enhance readability and match the website's aesthetics and branding.

Utilize web font services: Services like Google Fonts offer a wide selection of web-friendly fonts that can be easily integrated into your website using CSS.

```
body {

   font-family: Arial, sans-serif;

font-size: 16px;     line-height:

1.5;

}
```

Color Schemes:

Develop cohesive color schemes: Choose a color palette that reflects the website's branding and creates a visually harmonious experience for users.

```
body {

   background-color:          #f0f0f0;

color: #333;

} a

{

   color: #007bff;

}
```

Images and Multimedia:

Optimize and style images: Use CSS to adjust the size, position, borders, and effects of images to create a polished visual experience.

Implement responsive techniques: Ensure multimedia elements, such as videos and embedded content, are responsive and adapt well to different screen sizes and resolutions.

Navigation Design:

Design clear menus: Create navigation menus that are easy to understand and navigate, organizing content logically and providing clear labels for menu items.

Style navigation elements: Use CSS to style navigation menus, links, and buttons to improve usability and enhance the overall user experience**.**

Animations and Transitions:

Enhance user engagement: Incorporate subtle animations and transitions to add interactivity and visual interest to your website.

Apply CSS animations or transitions: Use CSS to create dynamic effects such as fading, sliding, or scaling elements to draw users' attention and improve engagement.

```
button {
   background-color: #007bff;
color: #fff;     border: none;
padding: 10px 20px;
cursor: pointer;
   transition: background-color 0.3s ease;
} button:hover
{
   background-color: #0056b3;
}
```

Test on different browsers: Ensure your website functions correctly and looks consistent across various web browsers by testing on popular browsers like Chrome, Firefox, Safari, and Edge.

Use prefixes or fall backs: Use vendor prefixes or provide fallbacks for CSS properties that may not be fully supported in older browsers to maintain compatibility.

Advantages of CSS:

- CSS saves a lot of time
- It helps to make consistent and spontaneous changes
- It improves the loading speed of the page
- CSS has the ability to re-position
- It has better device compatibility

Disadvantages of CSS:

- There could be cross-browser issues while using CSS
- There are multiple levels of CSS such as CSS, CSS 2, CSS 3. This can create confusion for non-developers and beginners.
- Lack of security is one of the major disadvantages of the Cascading style sheet.
- CSS has limited control over layouts. In order to achieve certain specific layout details like adding multiple columns to the table, other tools like Flexbox or Grid need to be used

BLOG PAGE

A blog site is a type of website or part of a website that is updated regularly with new content, typically in the form of articles or posts. These posts are often written in an informal or conversational style and can cover a wide range of topics, from personal experiences to professional advice in various fields. Blog sites can be maintained by individuals, companies, or groups of writers.

Characteristics of blog page

Regularly Updated Content

Blogs thrive on fresh content. Regular updates not only keep the audience engaged but also contribute to the site's visibility and relevance. This consistent addition of new content can reflect evolving perspectives, on-going developments in a particular field, or simply the blogger's latest interests or experiences. It's this dynamic nature that keeps readers returning, eager to see what's new.

Chronological Order

The reverse chronological order of posts means that the newest content appears at the top of the blog. This organization method ensures that readers always see the most recent posts first, making it easier for them to stay updated on the latest discussions, news, or insights without having to sift through older content. It mirrors the fast-paced, ever-evolving nature of the internet and people's desire for the latest information.

Comment Sections

Comments are a vital part of blogs, turning them from monologues into conversations. This interactive feature invites readers to engage with the content, share their thoughts, and contribute to the discussion. It creates a sense of community and can provide valuable feedback and new perspectives to both the blogger and other readers. Comments can enrich the content, challenge viewpoints, and foster a lively exchange of ideas.

Personal or Informal Style

Many blogs are characterized by their personal touch or informal approach, making complex topics more accessible or turning personal anecdotes into relatable stories. This style can make readers feel more connected to the blogger, as if they're having a conversation with a friend. It's this personal connection that often differentiates blogs from more traditional media, inviting readers into a shared space of exploration and discovery.

Variety of Topics

Blogs are incredibly diverse, covering every conceivable topic from daily life, hobbies, and personal growth, to specialized areas such as technology, education, finance, and beyond. This variety not only caters to a wide range of interests but also allows bloggers to niche down into specific areas, becoming go-to sources for particular types of information or insight. It reflects the vast tapestry of human interest and the internet's role in catering to it.

Multimedia Content

The integration of multimedia elements like images, videos, and links enriches blog content, making it more engaging and comprehensive. Visual aids can illustrate points more effectively than text alone, videos can provide in-depth explanations or demonstrations, and links can guide readers to additional resources for further exploration. This multimedia approach caters to different learning styles and keeps the content dynamic and interactive.

Archives

Archives are an essential feature, allowing readers to explore older content without navigating through each page manually. This accessibility to past posts can turn a blog into a valuable resource library on its topic(s) of focus. For new visitors or regular readers looking for specific information, archives offer an organized and efficient way to browse through the history of the blog, ensuring that valuable content remains accessible and doesn't get lost in the flow of new updates.

Types of blog pages

1. Personal Blog

Description: Personal blogs are online diaries or commentaries written by an individual, focusing on various aspects of their daily life, thoughts, and experiences. The content is often subjective and can cover a broad range of topics.

Example: Gretchen Rubin’s blog on happiness and habits, where she shares personal experiences and insights.

2. Collaborative Blog (Group Blog)

Description: These blogs involve multiple authors contributing posts, often around a unified theme or subject matter. Collaborative blogs can benefit from diverse viewpoints and expertise.

Example: Boing Boing is a collaborative blog with contributions from several writers on technology, culture, and more.

3. Micro blog

Description: Micro blogs are platforms for short-form content, allowing users to post brief text updates, images, or links. These blogs are great for quick, frequent updates.

Example: Twitter is a prime example, where individuals and organizations share updates, news, and thoughts in a concise format.

4. Video Blog (Vlog)

Description: Vlogs are blogs where the primary content is in video form. Vloggers often use platforms like YouTube to share their content, covering a wide range of topics from daily life to reviews.

Example: Casey Neistat, a popular YouTube, shares his life, tech reviews, and travel experiences through his vlogs.

5. Link Blog

Description: Link blogs focus on sharing curated links to external content, providing readers with resources and commentary on interesting articles, videos, and other online material.

Example: Kottke.org is a blog that curates fascinating links from around the web on various subjects, from technology to culture.

6. Photo Blog

Description: Photo blogs primarily feature photographic content. Photographers, both amateur and professional, use these platforms to showcase their work, tell stories through images, and share tips.

Example: Humans of New York started as a photo blog sharing portraits and stories of New York City's inhabitants.

7. Edu Blog

Description: Educational blogs focus on providing educational content, resources, and discussions. They can be used by teachers to supplement classroom instruction or by experts to share knowledge in a particular field.

Example: Edutopia offers a wealth of resources, insights, and inspiration for educators and students alike.

8. Business Blog

Description: Business blogs are maintained by companies and organizations to share industry insights, company news, and related content aimed at building a connection with their target audience and driving business goals.

Example: The Hub Spot Blog provides marketing, sales, and service tips and news, catering to professionals looking to improve their business strategies.

9. Niche Blog

Description: Niche blogs focus on a specific topic or interest area, providing detailed content and expertise. They cater to specific audiences looking for information or communities related to particular interests.

Example: Nomadic Matt specializes in budget travel, offering tips, guides, and personal experiences to help readers travel cheaper, longer, and smarter.

How to write blog page?

1. **Choose Your Topic**

Identify Your Audience: Understand who you are writing for. This will guide your topic choice, tone, and style.

Select a Topic: Choose something that is both of interest to your audience and for which you have enough knowledge or can research. Ensure it aligns with your blog's niche.

2. **Research**

Gather Information: Use reputable sources to gather information on your topic. This could include academic journals, expert blogs, and authoritative websites.

Keyword Research: Use tools like Google Keyword Planner or Ahrefs to find keywords that people are searching for related to your topic. This will help with SEO (Search Engine Optimization).

3. **Crafting Catchy Headlines**

Be Clear and Concise: Your headline should clearly communicate the benefit or value of reading your post. Avoid ambiguity.

Use Strong Keywords: Incorporate relevant keywords to improve SEO and attract more traffic. However, ensure they fit naturally and don't compromise readability.

Create Curiosity or Urgency: Phrases that evoke curiosity or a sense of urgency can encourage more clicks. For example, "5 Secrets to..." or "Don't Miss Out on..."

Use Numbers or Questions: Headlines with numbers (e.g., "10 Ways to...") or questions can be more engaging and direct.

4. **Ensuring a Well-Structured Central Idea**

Introduction: Start with a hook to grab attention and briefly outline what the reader will learn or how they will benefit from reading your post.

Body: Divide the content into sections with clear headings. Each section should cover a specific aspect of the topic, contributing to the development of your central idea.

Use Bullet Points or Lists: These can help break down information into digestible pieces, making it easier for readers to scan and understand.

Conclusion: Sum up the key points you've made, reinforce the central idea, and encourage action or further engagement from the reader.

5. **Edit and Proofread**

Grammar and Spelling: Use tools like Grammarly to check for grammatical errors and typos.

Readability: Ensure your sentences are clear and concise. Use tools like Hemingway Editor to check for readability.

Fact-check: Verify all your facts and figures and ensure all sources are credited appropriately. 6.

6. **Add Images and Multimedia**

Visual Appeal: Add relevant images, videos, or info graphics to make your post more engaging.

Optimization: Ensure your images are optimized for web use (compressed for faster loading without losing quality).

Create and maintain blog site

1. Define Blog Objectives

Purpose and Goals: Clearly defining what you hope to achieve with your blog, whether it's to share knowledge, drive product sales, or build a community, sets a clear direction for your efforts.

Target Audience and Topics: Understanding who your readers are allows you to tailor your content to their interests, needs, and challenges, making your blog more relevant and engaging.

2. Choose a Blogging Platform

Selection: WordPress, Blogger, and Medium are popular choices, each with its unique advantages. WordPress, for example, offers extensive customization but requires hosting. Medium provides a built-in audience at the cost of less personalization.

Factors to Consider: Look into how user-friendly the platform is, what customization options it offers, and whether it can grow with your blog over time.

3. Design a User-Friendly Interface

Layout: The design should be clean and intuitive, prioritizing easy navigation to help readers find content.

Design Elements: A visually appealing blog with a readable font and coherent color scheme enhances user experience and keeps visitors engaged.

4. Content Strategy

Theme Alignment: Your content should reflect the overarching theme of your blog, providing depth and value in your niche.

Posting Schedule: Regular updates keep your audience engaged and improve SEO rankings.
Plan your posts to maintain consistency.

Pages in Blogger:

Static content containers within a blog.

Used for permanent content like About Me/Us, Contact, Privacy Policy.

Accessible from the main navigation menu.

Ideal for information not changing frequently.

6. Stats in Blogger:

Statistical reports on blog performance.

Metrics include page views, visitors, traffic sources, popular content, demographics.

Helps understand audience engagement and content effectiveness.

Guides content strategy and optimization efforts.

7. SEO Optimization

On-Page SEO: Implementing keywords, meta tags, and descriptive URLs helps search engines understand and rank your content.

Best Practices: Staying up-to-date with SEO strategies is crucial for increasing visibility and driving organic traffic to your blog.

8. Multimedia Integration

Content Enrichment: Images, info graphics, and videos can break up text, illustrate points, and engage readers more deeply with your content.

Optimization: Ensure multimedia elements are optimized for fast loading and are accessible to all users.

9. Social Media Integration

Sharing Buttons: Easy-to-use social sharing options encourage readers to share your content, increasing its reach.

Platform Integration: Connecting your blog with your social media profiles can drive traffic and enhance your online presence.

10. Commenting System

Engagement: A robust commenting system invites reader interaction, increasing engagement and providing valuable feedback.

Moderation: Moderating comments helps maintain a respectful and constructive community dialogue.

11. Responsive Design

Cross-Device Accessibility: A responsive design ensures your blog is accessible and visually appealing across all devices, essential for reaching a wider audience.

User Experience Testing: Regularly test your blog on various devices to ensure the user experience remains consistent and functional.

12. Analytics Integration

Traffic Insights: Analytics tools like Google Analytics offer insights into your blog's performance, visitor behaviours, and traffic sources.

Content Refinement: Use these insights to refine your strategy, focusing on what works best for your audience.

13. Security Measures

Protection: Implementing security best practices protects your blog from hacking, malware, and other cyber threats.

Updates: Keeping your blogging software and plugins updated is crucial for security and functionality.

14. Monetization Strategies

Revenue Generation: Explore various monetization options like advertisements, sponsored content, and affiliate marketing suited to your blog's niche and audience.

Integration: Carefully integrate monetization elements to ensure they don't detract from the user experience.

15. Regular Updates and Maintenance

Platform and Plugin Updates: Regularly updating your blogging platform and plugins ensures your site runs smoothly and remains secure.

Content Review: Periodically review your content for relevancy, updating or archiving posts as necessary.

16. Community Building

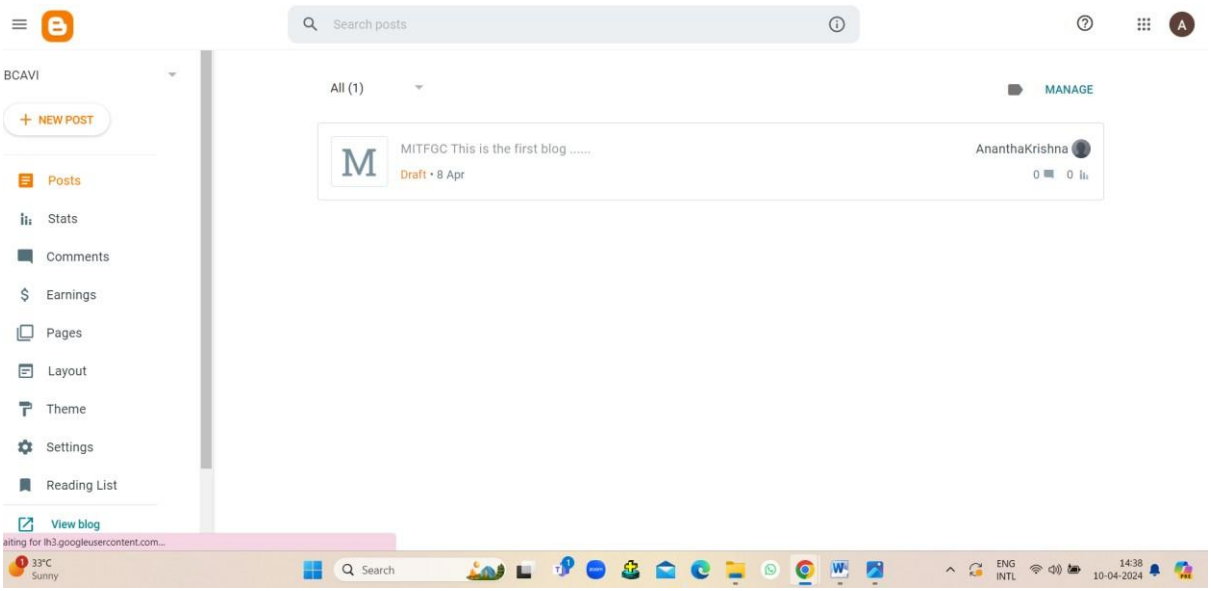
Engagement: Actively engaging with your readers through comments, social media, and other channels helps build a loyal community around your blog.

Participation: Join relevant discussions and forums to increase your visibility and establish your blog within its niche community.

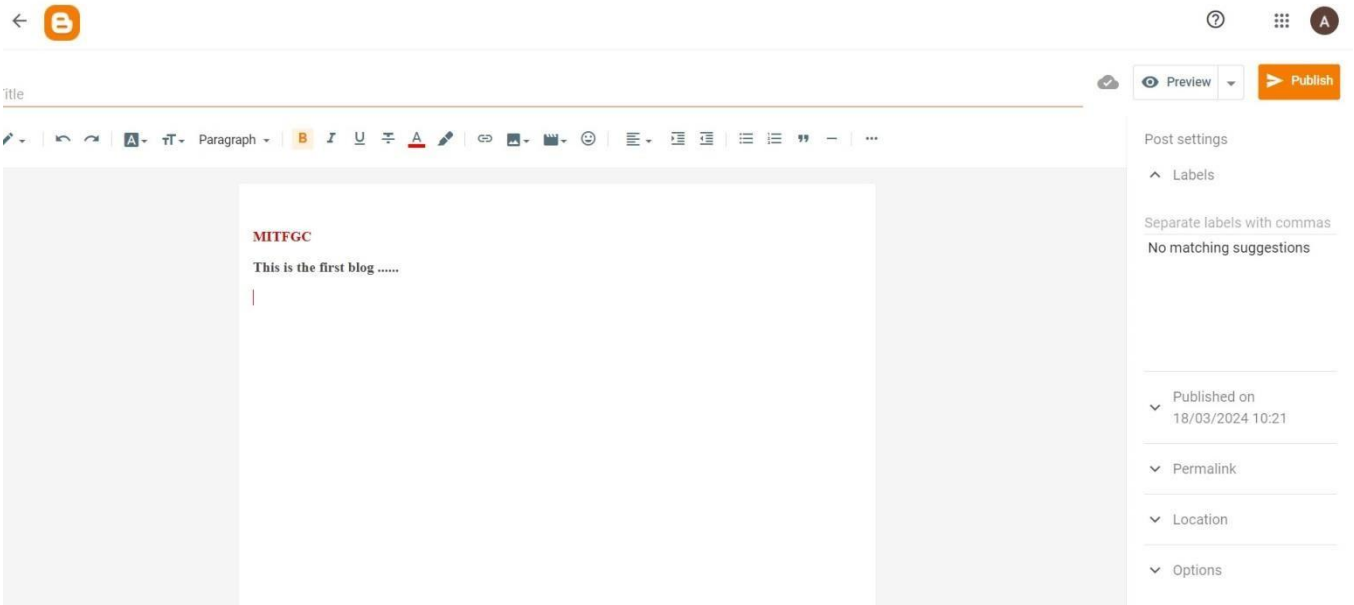
17. Backup and Recovery

Data Protection: Regular backups protect against data loss, ensuring your content is safe in case of technical issues.

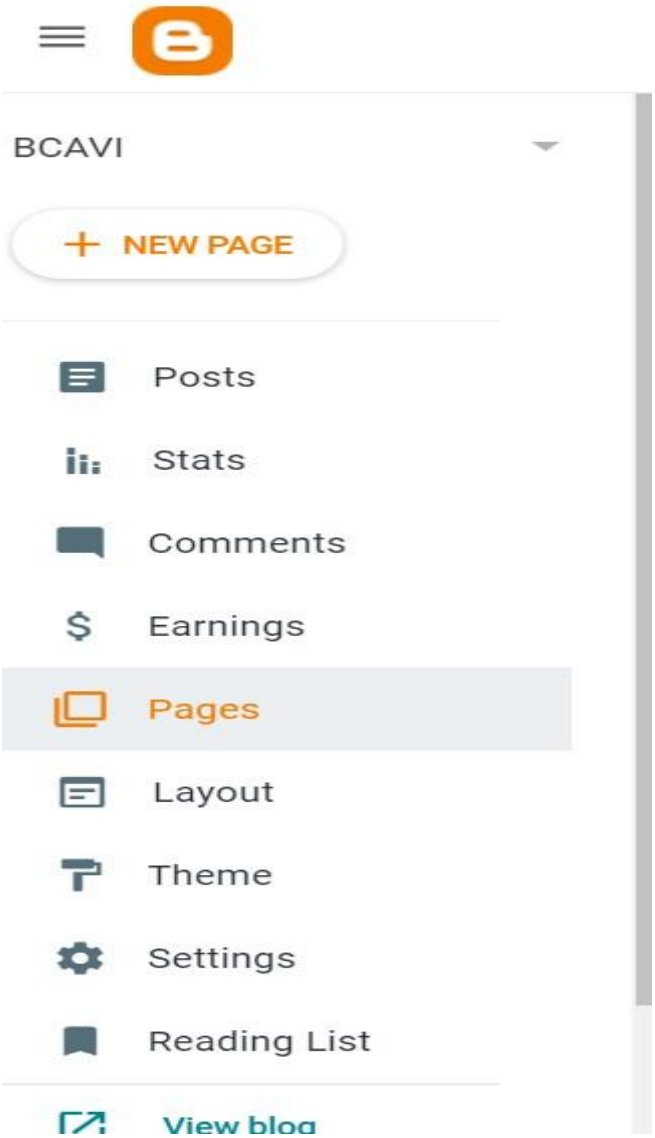
Recovery Plan: Have a clear recovery plan in place to quickly restore your blog if needed.



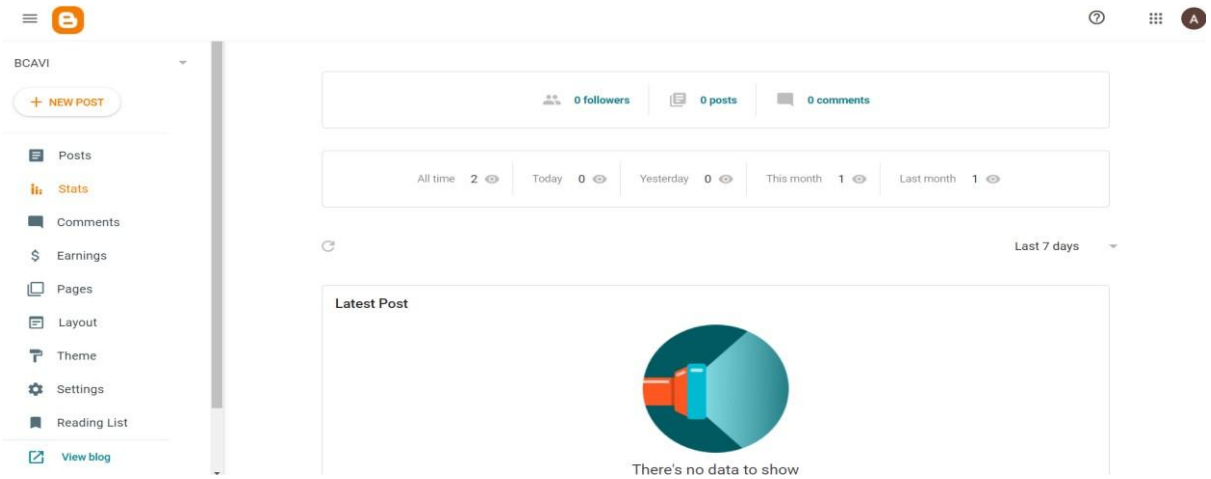
Home Page of blog



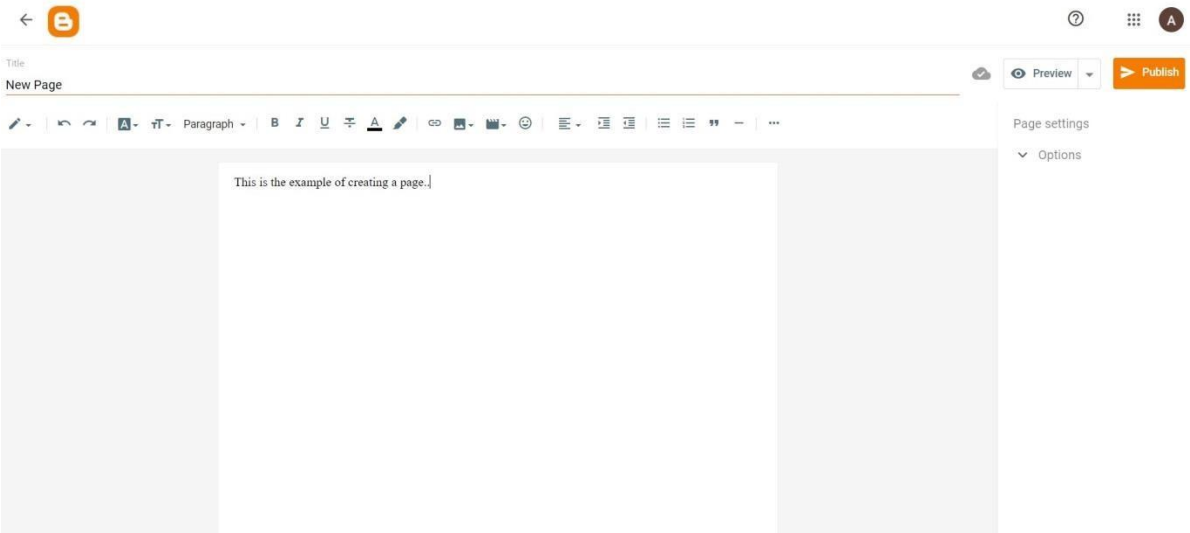
Writing a blog page



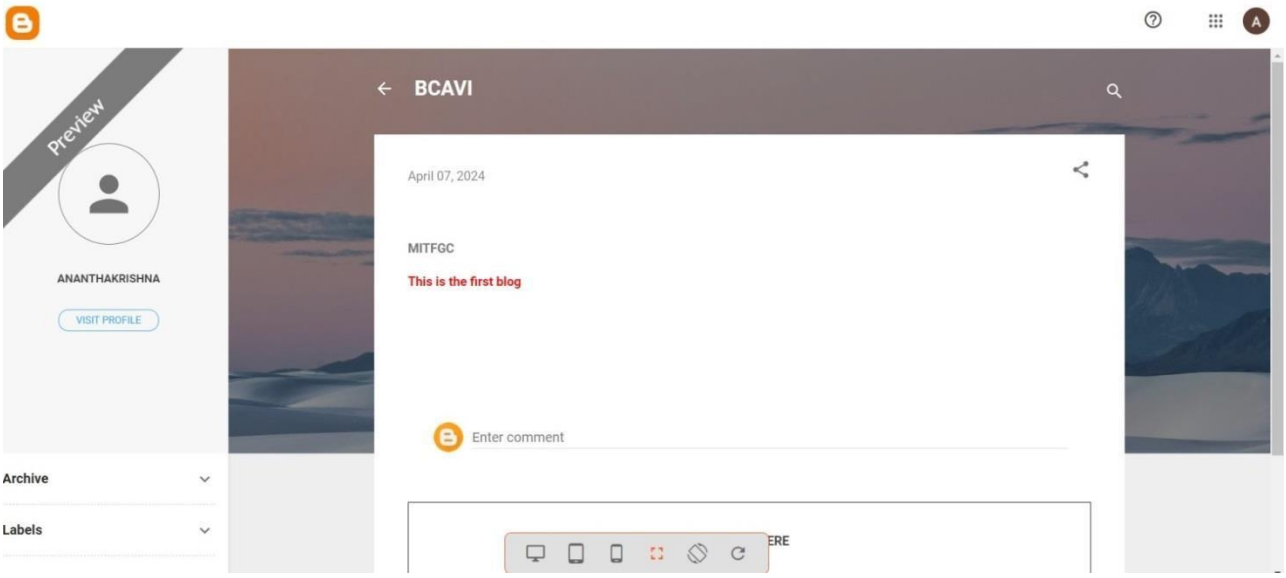
Options we get in blog page



Stats page of the blog



Creating a new page on the blog page



View of a simple blog page